



Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
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Laporan Akhir

Praktikum Jaringan Komputer

VLAN, Trunking, dan OSPF Multi-Vendor

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1 Langkah-Langkah Percobaan

Praktikum 1 Konfigurasi Cisco Switch

1. Masuk ke Terminal Cisco Switch

```
1 Switch> enable
2 Switch# configure terminal
3 Switch(config)#
```

2. Membuat VLAN

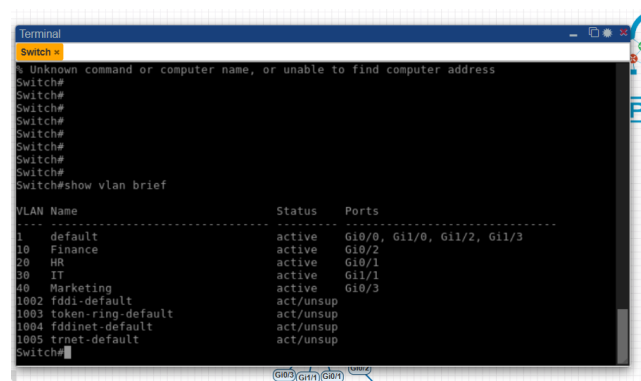
```
1 vlan 10
2   name Finance
3 exit
4
5 vlan 20
6   name HR
7 exit
8
9 vlan 30
10  name IT
11 exit
12
13 vlan 40
14  name Marketingg
15 exit
```

Verifikasi:

```
1 show vlan brief
```

3. Pastikan VLAN Sudah Dibuat dan Statusnya Aktif

Hasil VLAN.



Gambar 1: Show VLAN brief

4. Konfigurasi Access Port untuk Setiap VLAN

Port Access harus disesuaikan dengan interface pada Cisco Switch.

```
1 Switch#conf t
2
3 interface gi0/2
4   description PC-FINANCE-VLAN10
5   switchport mode access
6   switchport access vlan 10
7   no shutdown
8 exit
9
10 interface gi0/1
11   description PC-HR-VLAN20
12   switchport mode access
13   switchport access vlan 20
14   no shutdown
15 exit
16
17 interface gi1/1
18   description PC-IT-VLAN30
19   switchport mode access
20   switchport access vlan 30
21   no shutdown
22 exit
23
24 interface gi0/3
25   description PC-MARKETING-VLAN40
26   switchport mode access
27   switchport access vlan 40
28   no shutdown
29 exit
```

5. Konfigurasi Koneksi Trunk antara Switch dan Router

Agar lalu lintas dari beberapa VLAN dapat diteruskan ke Cisco Router, interface yang menghubungkan switch dengan router dikonfigurasi dalam mode trunk. VLAN yang diizinkan melewati koneksi ini meliputi VLAN 10, 20, 30, dan 40.

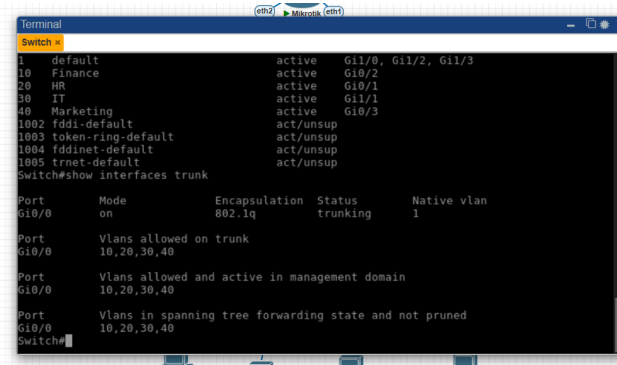
Port Switch	Perangkat Tujuan	Tipe Port	VLAN Aktif
Gi0/0	Cisco Router Gi0/2	Trunk	10, 20, 30, 40

Konfigurasi:

```
1 Switch#conf t
2
3 interface gi0/0
4   switchport trunk encapsulation dot1q
5   description TRUNK-T0-CISCO-ROUTER
6   switchport mode trunk
7   switchport trunk allowed vlan 10,20,30,40
8   no shutdown
9 exit
```

Verifikasi:

```
1 show interfaces trunk
```



Gambar 2: show interfaces trunk

Save semua konfigurasi:

```
1 write memory
```

Praktikum 2 Konfigurasi Cisco Router

1. Masuk ke Cisco Router

```
1 enable
2 configure terminal
```

2. Konfigurasi Interface Trunk ke Switch

Pilih interface yang terhubung ke Cisco Switch. Pada topologi, interface yang terhubung adalah Gig0/2.

```
1 interface gig0/2
2 description TRUNK-T0-CISCO-SWITCH
3 no ip address
4 no shutdown
5 exit
```

3. Konfigurasi Subinterface VLAN 10

```
1 interface gig0/2.10
2 description GATEWAY-VLAN10-FINANCE
3 encapsulation dot1Q 10
4 ip address 192.168.10.1 255.255.255.0
5 no shutdown
6 exit
```


4. Konfigurasi Subinterface VLAN 20

```
1 interface gig0/2.20
2   description GATEWAY-VLAN20-HR
3   encapsulation dot1Q 20
4   ip address 192.168.20.1 255.255.255.0
5   no shutdown
6   exit
```

5. Konfigurasi Subinterface VLAN 30

```
1 interface gig0/2.30
2   description GATEWAY-VLAN30-IT
3   encapsulation dot1Q 30
4   ip address 192.168.30.1 255.255.255.0
5   no shutdown
6   exit
```

6. Konfigurasi Subinterface VLAN 40

```
1 interface gig0/2.40
2   description GATEWAY-VLAN40-MARKETING
3   encapsulation dot1Q 40
4   ip address 192.168.40.1 255.255.255.0
5   no shutdown
6   exit
```

7. Konfigurasi DHCP-Server untuk VLAN 10

```
1 ip dhcp excluded-address 192.168.10.1 192.168.10.10
2
3 ip dhcp pool VLAN10-FINANCE
4   network 192.168.10.0 255.255.255.0
5   default-router 192.168.10.1
6   dns-server 8.8.8.8
7   exit
```

8. Konfigurasi DHCP-Server untuk VLAN 20

```
1 ip dhcp excluded-address 192.168.20.1 192.168.20.10
2
3 ip dhcp pool VLAN20-HR
4   network 192.168.20.0 255.255.255.0
5   default-router 192.168.20.1
6   dns-server 8.8.8.8
7   exit
```

9. Konfigurasi IP Static pada VPCS untuk VLAN 30 dan 40

PC VLAN 30:

```
1 ip 192.168.30.10/24 192.168.30.1
```

PC VLAN 40:

```
1 ip 192.168.40.10/24 192.168.40.1
```

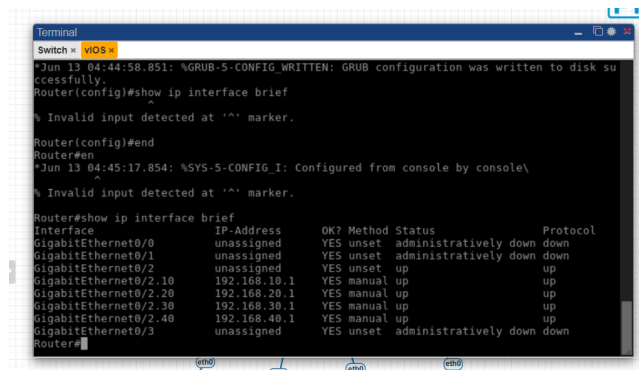
PC VLAN 10 dan 20 menggunakan DHCP:

```
1 ip dhcp
```

10. Verifikasi Cisco Router

Cek interface:

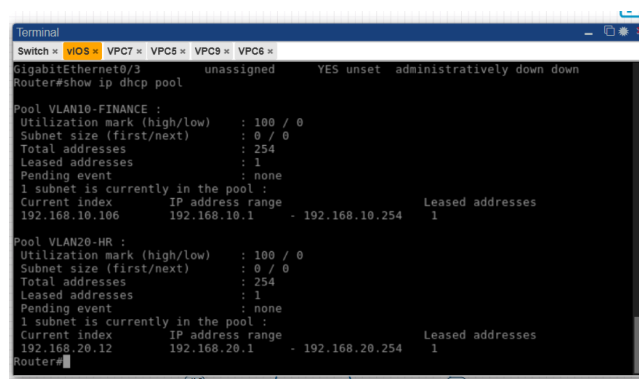
```
1 show ip interface brief
```



Gambar 3: Show IP interface brief

Cek DHCP Pool:

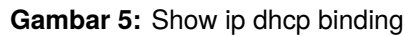
```
1 show ip dhcp pool
```



Gambar 4: Show ip dhcp pool

Cek client DHCP yang mendapatkan IP:

```
1 show ip dhcp binding
```



Masuk ke PC VLAN 10 untuk request IP secara DHCP:

```

Terminal
Switch > vIOS > VPC7 > VPC8 > VPC9 > VPC4 >
Press '?' to get help.

VPCS>
VPCS> ip dhcp
DORA IP 192.168.10.105/24 GW 192.168.10.1

VPCS> ip dhcp
DORA IP 192.168.10.105/24 GW 192.168.10.1

VPCS> show ip

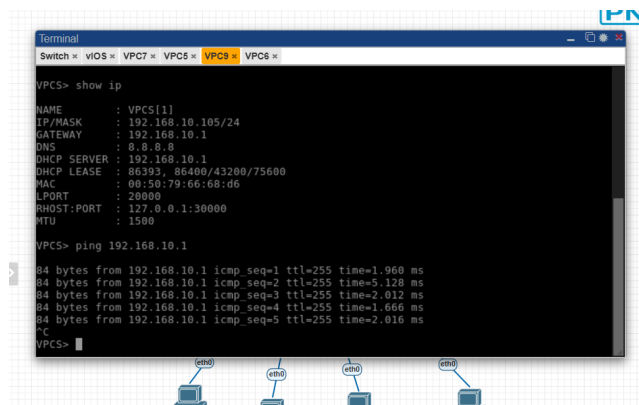
NAME      : VPCS[1]
IP/MASK   : 192.168.10.105/24
GATEWAY   : 192.168.10.1
DNS       : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE : 86393, 86400/43200/75600
MAC       : 00:50:79:66:68:d6
PORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VPCS>

```

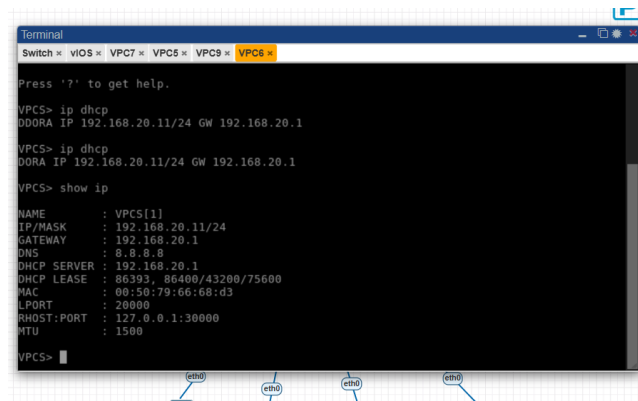
Gambar 6: request ip dhcp

```
1 ping 192.168.10.1
```



Gambar 7: ping gateway VLAN 10

6

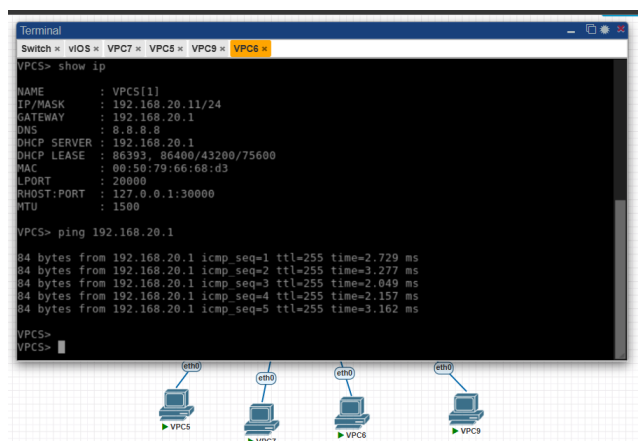


Gambar 8: request ip dhcp

```
1 ip dhcp
2 show ip
```

Ping gateway VLAN 20:

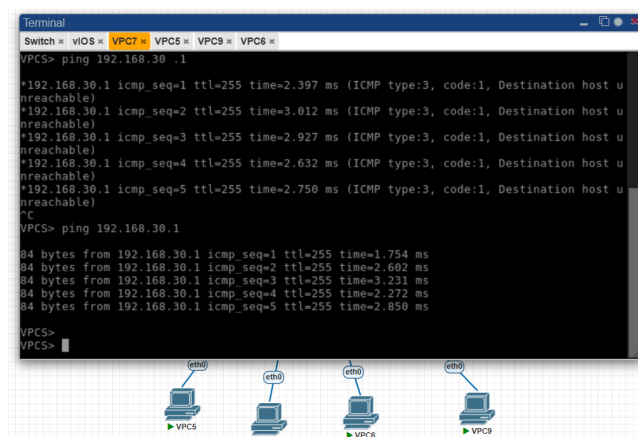
```
1 ping 192.168.20.1
```



Gambar 9: ping gateway VLAN 20

Ping gateway VLAN 30:

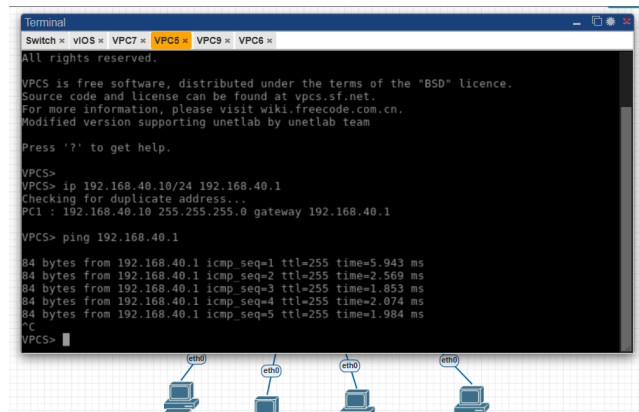
```
1 ping 192.168.30.1
```



Gambar 10: ping gateway vlan 30

Ping gateway VLAN 40:

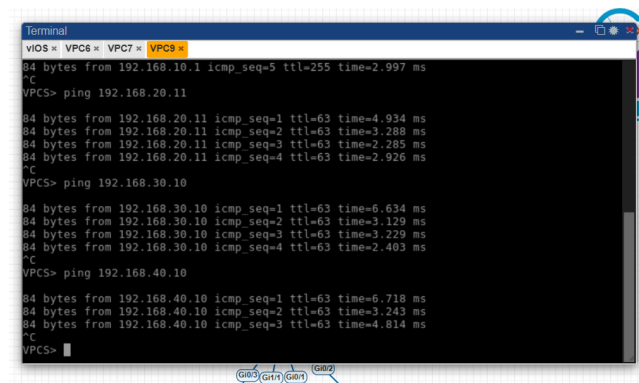
```
1 ping 192.168.40.1
```



Gambar 11: ping gateway vlan 40

Uji ping antar VLAN dari VLAN 10:

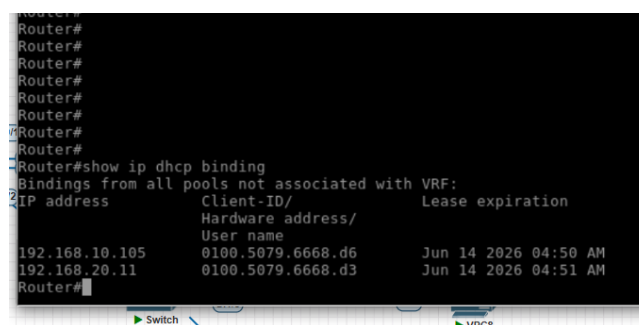
```
1 ping 192.168.20.11
2 ping 192.168.30.10
3 ping 192.168.40.10
```



Gambar 12: dari vlan 10

Lihat DHCP Binding:

```
1 show ip dhcp binding
```



Gambar 13: ip dhcp binding

Amati IP yang diberikan oleh router ke client sesuai dengan IP pada PC VLAN 10 dan VLAN 20.

12. Hasil yang Diharapkan

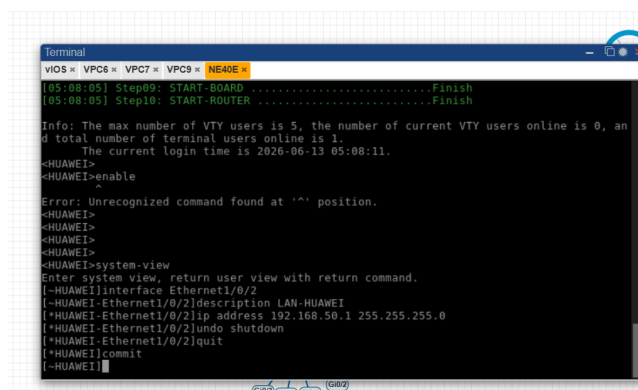
- Cisco Router memiliki subinterface VLAN 10, 20, 30, dan 40.
- VLAN 10 mendapat gateway 192.168.10.1.
- VLAN 20 mendapat gateway 192.168.20.1.
- VLAN 30 mendapat gateway 192.168.30.1.
- VLAN 40 mendapat gateway 192.168.40.1.
- PC VLAN 10 dan VLAN 20 mendapatkan IP dari DHCP.
- Semua PC dapat ping gateway masing-masing.
- PC antar-VLAN dapat saling ping melalui Cisco Router.

Praktikum 3 Konfigurasi Huawei Router dan IP Address Antar Router

1. Konfigurasi LAN Huawei

Konfigurasi IP Address untuk LAN Huawei:

```
1 system-view
2
3 interface Ethernet1/0/2
4   description LAN-HUAWEI
5   ip address 192.168.50.1 255.255.255.0
6   undo shutdown
7   quit
8
9 commit
```



Gambar 14

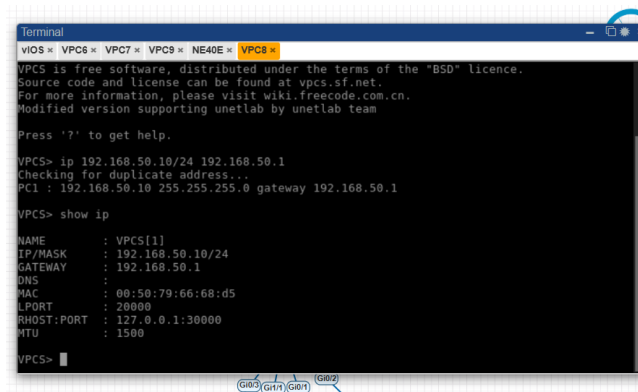
2. Konfigurasi IP VPC LAN Huawei

Pada PC LAN Huawei:

```
1 ip 192.168.50.10/24 192.168.50.1
```

Verifikasi:

```
1 show ip
```



Gambar 15

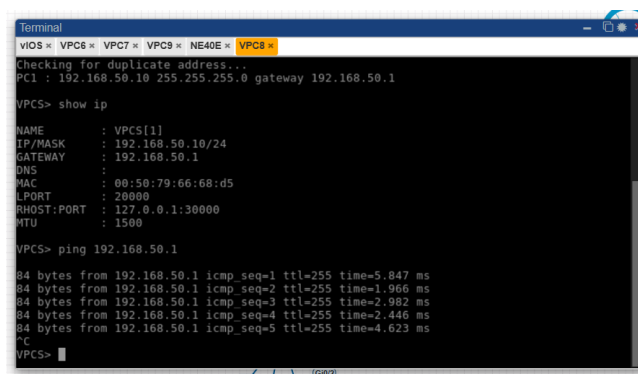
Testing ping gateway Huawei:

```
1 ping 192.168.50.1
```

Pastikan hasilnya reply.

Target:

PC LAN Huawei dapat melakukan ping ke gateway 192.168.50.1.



Gambar 16

3. Konfigurasi IP Link Cisco Huawei Link 1

Network:

```
1 10.10.10.0/30
```

Perangkat	Interface	IP Address
Cisco Router	Gi0/3	10.10.10.1/30
Huawei Router	Ethernet1/0/0	10.10.10.2/30

Cisco Router:

```
1 enable
2 configure terminal
3
4 interface gi0/3
```

```

5 description LINK1-T0-HUAWEI
6 ip address 10.10.10.1 255.255.255.252
7 no shutdown
8 exit

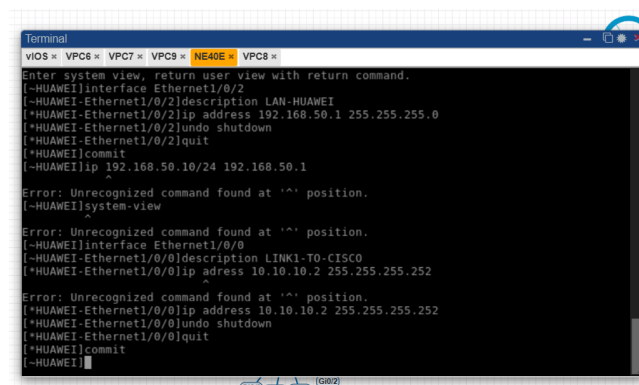
```

Huawei Router:

```

1 system-view
2
3 interface Ethernet1/0/0
4 description LINK1-T0-CISCO
5 ip address 10.10.10.2 255.255.255.252
6 undo shutdown
7 quit
8
9 commit

```



Gambar 17

1.1 4. Konfigurasi IP Link Cisco Huawei Link 2

Network:

```

1 10.10.11.0/30

```

Perangkat	Interface	IP Address
Cisco Router	Gi0/0	10.10.11.1/30
Huawei Router	Ethernet1/0/4	10.10.11.2/30

Cisco Router:

```

1 interface gi0/0
2 description LINK2-T0-HUAWEI
3 ip address 10.10.11.1 255.255.255.252
4 no shutdown
5 exit

```



```

Terminal
VPCS VPC6 VPC7 VPC8 NE40E VPC8
Router(config)#ex
Router(config)#end
Router#
*Jun 13 05:16:23.495: %SYS-5-CONFIG_I: Configured from console by console
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g10/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#ex
Router(config)#
*Jun 13 05:17:09.971: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 13 05:17:10.971: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
*Jun 13 05:17:16.938: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun 13 05:17:17.643: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully

```

Gambar 18

Huawei Router:

```

1 system-view
2
3 interface Ethernet1/0/4
4   description LINK2-T0-CISCO
5   ip address 10.10.11.2 255.255.255.252
6   undo shutdown
7   quit
8
9 commit

```

```

Terminal
VPCS VPC6 VPC7 VPC8 NE40E VPC8
Error: Unrecognized command found at '^' position.
[~HUAWEI]system-view
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[~HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
^
Error: Unrecognized command found at '^' position.
[~HUAWEI-Ethernet1/0/0]ip address 10.10.2 255.255.255.252
[~HUAWEI-Ethernet1/0/0]undo shutdown
[~HUAWEI-Ethernet1/0/0]quit
[~HUAWEI]commit
[~HUAWEI]system-view
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[~HUAWEI-Ethernet1/0/4]ip address 10.10.11.2 255.255.255.252
[~HUAWEI-Ethernet1/0/4]undo shutdown
[~HUAWEI-Ethernet1/0/4]quit
[~HUAWEI]commit
[~HUAWEI]

```

Gambar 19

5. Konfigurasi IP Link Huawei MikroTik

Network:

```

1 10.10.20.0/30

```

Perangkat	Interface	IP Address
Huawei Router	Ethernet1/0/3	10.10.20.1/30
MikroTik	ether1	10.10.20.2/30

Huawei Router:

```

1 system-view
2
3 interface Ethernet1/0/3

```

```

4 description LINK-T0-MIKROTIK
5 ip address 10.10.20.1 255.255.255.252
6 undo shutdown
7 quit
8
9 commit

```

MikroTik:

```

1 /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI

```

6. Konfigurasi IP Link Cisco MikroTik

Network:

```

1 10.10.30.0/30

```

Perangkat	Interface	IP Address
Cisco Router	Gi0/1	10.10.30.1/30
MikroTik	ether2	10.10.30.2/30

Cisco Router:

```

1 interface gi0/1
2 description LINK-T0-MIKROTIK
3 ip address 10.10.30.1 255.255.255.252
4 no shutdown
5 exit

```

MikroTik:

```

1 /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO

```

7. Verifikasi Interface

Cisco Router:

```

1 show ip interface brief

```

Pastikan IP pada interface berikut sudah aktif.

```
Terminal
vIOS < VPC6 < VPC7 < VPC8 < NE40E < VPC8 < Mikrotik <
Do you want to see the software license? [Y/n]: n
[?] Gives the list of available commands
command [?] Gives help on the command and list of arguments
[Tab] Completes the command/word. If the input is ambiguous,
a second [Tab] gives possible options
/ Move up to base level
.- Move up one level
/command Use command at the base level

Change your password
[admin@mikrotik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CI
[admin@mikrotik] > ow (line 1 column 1)ief
[admin@mikrotik] >
[admin@mikrotik] >
[admin@mikrotik] >
[admin@mikrotik] >
[admin@mikrotik] >
[admin@mikrotik] >
[admin@mikrotik] >
[admin@mikrotik] >
[admin@mikrotik] > show ip interface brief
bad command name show (line 1 column 1)
[admin@mikrotik] > /ip address print
# ADDRESS NETWORK INTERFACE
0 ::: TO-HUAMEI 10.10.20.2/30 10.10.20.0 ether1
1 ::: TO-CISCO 10.10.30.2/30 10.10.30.0 ether2
[admin@mikrotik] >
```

Gambar 20

Huawei Router:

```
1 display ip interface brief
```

Pastikan IP sudah sesuai pada interface dan statusnya up.

```
Terminal
vIOS < VPC6 < VPC7 < VPC8 < NE40E < VPC8 < Mikrotik <
Router(config)#
Router(config)#
Router(config)#interface gi0/1
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#ex
Router(config)#
*Jun 13 05:22:14.889: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 13 05:22:15.889: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config)#
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
*Jun 13 05:22:27.986: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun 13 05:22:28.688: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Router(config)#end
Router#sho
*Jun 13 05:24:14.741: %SYS-5-CONFIG_I: Configured from console by console
Type "show ?" for a list of subcommands
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up up
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual up up
Router#
```

Gambar 21

MikroTik:

```
1 /ip address print
```

Pastikan IP sudah sesuai dengan konfigurasi.

```

Terminal
vIOS x VPC6 x VPC7 x VPC8 x NE40E x VPC8 x Mikrotik x
Do you want to see the software license? [Y/n]: n
[?] Gives the list of available commands
command [?] Gives help on the command and list of arguments
[Tab] Completes the command/word. If the input is ambiguous,
a second [Tab] gives possible options
/ Move up to base level
./ Move up one level
/command Use command at the base level

Change your password
[admin@Mikrotik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CI
[admin@Mikrotik] > ow (line 1 column 1)ief
[admin@Mikrotik] >
[admin@Mikrotik] >
[admin@Mikrotik] >
[admin@Mikrotik] >
[admin@Mikrotik] >
[admin@Mikrotik] >
[admin@Mikrotik] >
[admin@Mikrotik] >
[admin@Mikrotik] > show ip interface brief
bad command name show (line 1 column 1)
[admin@Mikrotik] > /ip address print
# ADDRESS NETWORK INTERFACE
0 :::: TO-HUAWEI 10.10.20.2/30 10.10.20.0 ether1
1 :::: TO-CISCO 10.10.30.2/30 10.10.30.0 ether2
[admin@Mikrotik] >

```

Gambar 22

8. Verifikasi Ping Antar Router

Dari Cisco Router:

```

1 ping 10.10.10.2
2 ping 10.10.11.2
3 ping 10.10.30.2

```

Target:

Cisco harus bisa ping ke Huawei dan MikroTik.

```

Terminal
vIOS x VPC6 x VPC7 x VPC8 x NE40E x VPC8 x Mikrotik x
sk. Please wait...
*Jun 13 05:22:28.688: %GRUB-5-CONFIG WRITTEN: GRUB configuration was written to disk su
ccessfully.
Router(config)#end
Router#sho
*Jun 13 05:24:14.741: %SYS-5-CONFIG I: Configured from console by consolew
* Type "show ?" for a list of subcommands
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up
GigabitEthernet0/1 10.10.30.1 YES manual up
GigabitEthernet0/2 unassigned YES unset up
GigabitEthernet0/2.10 192.168.10.1 YES manual up
GigabitEthernet0/2.20 192.168.20.1 YES manual up
GigabitEthernet0/2.30 192.168.30.1 YES manual up
GigabitEthernet0/2.40 192.168.40.1 YES manual up
GigabitEthernet0/3 10.10.10.1 YES manual up
Router#
Router#
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/4 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router#

```

Gambar 23

Dari Huawei Router:

```

1 ping 10.10.10.1
2 ping 10.10.11.1
3 ping 10.10.20.2

```

Target:

Huawei harus bisa ping ke Cisco dan MikroTik.

```

Terminal
vIOS < VPC6 < VPC7 < VPC9 < NE40E < VPC8 < Mikrotik <
Ethernet1/0/8.4094      128.1.138.137/16      up      up      l3vpn
Ethernet1/0/9           unassigned            up      down   --
GigabitEthernet0/0/0    128.1.138.137/16      up      up      l3vpn
LoopBack2147483647      128.1.138.137/16      up      up(s)   l3vpn
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=4 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=1 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=1 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=1 ms
--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/4 ms
[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms
--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms
[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms
--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break

```

Gambar 24

Dari MikroTik:

- 1 ping 10.10.20.1
- 2 ping 10.10.30.1

Target:

MikroTik harus bisa ping ke Cisco dan Huawei.

```

Terminal
vIOS < VPC6 < VPC7 < VPC9 < NE40E < VPC8 < Mikrotik <
Change your password
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CI
[admin@MikroTik] > ow (line 1 column 1)ief
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] > show ip interface brief
add command name show (line 1 column 1)
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 :::: TO-HUAWEI
1 10.10.20.2/30 10.10.20.0 ether1
1 :::: TO-CISCO
1 10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 2ms
1 10.10.20.1 56 255 1ms
2 10.10.20.1 56 255 1ms
3 10.10.20.1 56 255 1ms
4 10.10.20.1 56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 1ms
1 10.10.20.1 56 255 1ms
sent=2 received=2 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=1ms
[admin@MikroTik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 1ms
2 10.10.30.1 56 255 1ms
3 10.10.30.1 56 255 1ms
4 10.10.30.1 56 255 2ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
[admin@MikroTik] >

```

Gambar 25

Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Implementasi OSPF pada Topologi Jaringan

Topologi yang digunakan memiliki dua koneksi langsung antara Cisco Router dan Huawei Router yang berperan sebagai jalur utama komunikasi. Selain itu, MikroTik diintegrasikan sebagai jalur redundan yang akan digunakan apabila kedua koneksi utama mengalami gangguan.

Koneksi	Cost OSPF	Peran
Cisco – Huawei (Link 1)	10	Jalur prioritas utama
Cisco – Huawei (Link 2)	10	Jalur prioritas utama
Cisco – MikroTik	100	Jalur redundansi
Huawei – MikroTik	100	Jalur redundansi

2. Advertised Network pada OSPF

Network yang diumumkan ke dalam proses routing OSPF berbeda pada setiap router sesuai dengan interface dan segmen jaringan yang dimilikinya.

Cisco Router

Prefix Jaringan	Deskripsi
10.10.10.0/30	Koneksi menuju Huawei (Link 1)
10.10.11.0/30	Koneksi menuju Huawei (Link 2)
10.10.30.0/30	Koneksi menuju MikroTik
192.168.10.0/24	Segmen VLAN 10 (Finance)
192.168.20.0/24	Segmen VLAN 20 (HR)
192.168.30.0/24	Segmen VLAN 30 (IT)
192.168.40.0/24	Segmen VLAN 40 (Marketing)

Huawei Router

Prefix Jaringan	Deskripsi
10.10.10.0/30	Koneksi menuju Cisco (Link 1)
10.10.11.0/30	Koneksi menuju Cisco (Link 2)
10.10.20.0/30	Koneksi menuju MikroTik
192.168.50.0/24	Jaringan lokal Huawei

MikroTik

Prefix Jaringan	Deskripsi
10.10.20.0/30	Koneksi ke Huawei Router
10.10.30.0/30	Koneksi ke Cisco Router

3. Konfigurasi OSPF pada Cisco Router

Masuk ke Cisco Router:

```
1 enable
2 configure terminal
```

Konfigurasi OSPF:

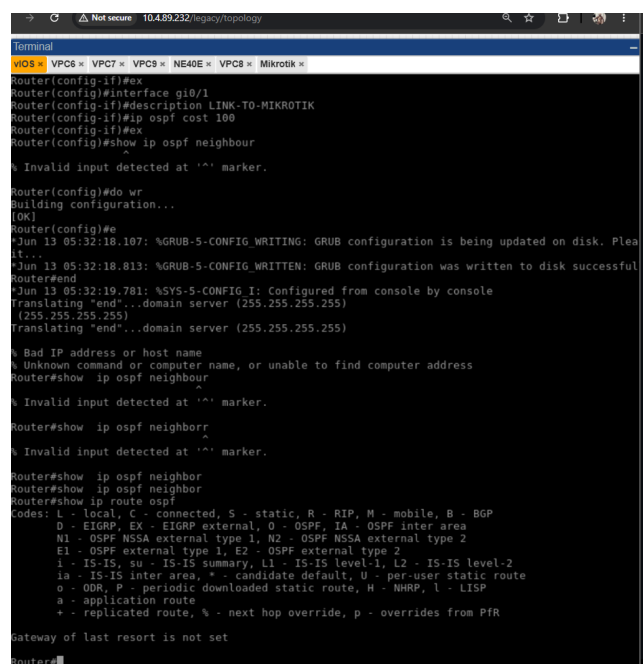
```
1 router ospf 1
2   router-id 1.1.1.1
3   network 10.10.10.0 0.0.0.3 area 0
4   network 10.10.11.0 0.0.0.3 area 0
5   network 10.10.30.0 0.0.0.3 area 0
6   network 192.168.10.0 0.0.0.255 area 0
7   network 192.168.20.0 0.0.0.255 area 0
8   network 192.168.30.0 0.0.0.255 area 0
9   network 192.168.40.0 0.0.0.255 area 0
10 exit
```

Atur OSPF Cost:

```
1 interface gi0/3
2   description LINK1-T0-HUAWEI
3   ip ospf cost 10
4 exit
5
6 interface gi0/0
7   description LINK2-T0-HUAWEI
8   ip ospf cost 10
9 exit
10
11 interface gi0/1
12  description LINK-T0-MIKROTIK
13  ip ospf cost 100
14 exit
```

Verifikasi:

```
1 show ip ospf neighbor
2 show ip route ospf
3 show ip protocols
```



```
Terminal
VPCS VPC7 VPC8 NE40E VPC8 Mikrotik
Router(config-if)#ex
Router(config)#interface gi0/1
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip ospf cost 100
Router(config-if)#ex
Router(config)#show ip ospf neighbour
% Invalid input detected at '^' marker.
Router(config)#do wr
Building configuration...
[OK]
Router(config)#e
Jun 13 05:32:18.107: %GRUB-5-CONFIG WRITING: GRUB configuration is being updated on disk. Plea
it...
Jun 13 05:32:18.813: %GRUB-5-CONFIG WRITTEN: GRUB configuration was written to disk successful
Router#end
Jun 13 05:32:19.781: %SYS-5-CONFIG I: Configured from console by console
Translating "end"...domain server (255.255.255.255)
(255.255.255.255)
Translating "end"...domain server (255.255.255.255)
% Bad IP address or host name
% Unknown command or computer name, or unable to find computer address
Router#show ip ospf neighbour
% Invalid input detected at '^' marker.
Router#show ip ospf neighborr
% Invalid input detected at '^' marker.
Router#show ip ospf neighbor
Router#show ip ospf neighbor
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set
Router#
```

Gambar 26

```

Terminal
VOS > VPC6 > VPC7 > VPC8 > NE40E > VPC8 > Mikrotik >
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
    Routing Information Sources:
      Gateway         Distance         Last Update
    Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.20.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

Router#
Router#
Router#
Router#
Router#

```

Gambar 27

Jika IP VLAN tidak muncul:

```

1 wr
2 reload

```

1.2 4. Konfigurasi OSPF pada Huawei Router

```

1 system-view
2
3 ospf 1 router-id 2.2.2.2
4 area 0.0.0.0
5   network 10.10.10.0 0.0.0.3
6   network 10.10.11.0 0.0.0.3
7   network 10.10.20.0 0.0.0.3
8   network 192.168.50.0 0.0.0.255
9 quit
10 quit

```

Atur OSPF Cost:

```

1 interface Ethernet1/0/0
2   description LINK1-T0-CISCO
3   ospf cost 10
4   quit
5
6 interface Ethernet1/0/4
7   description LINK2-T0-CISCO
8   ospf cost 10
9   quit
10
11 interface Ethernet1/0/3
12   description LINK-T0-MIKROTIK
13   ospf cost 100
14   quit

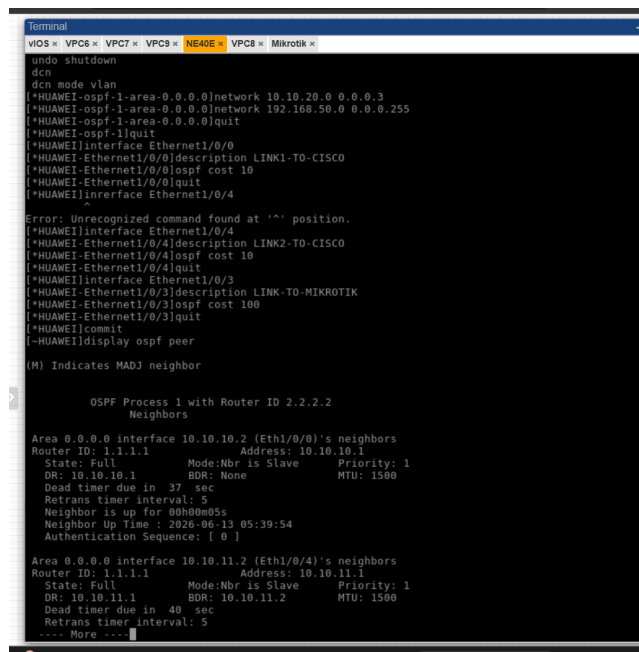
```



```
15
16 commit
```

Verifikasi:

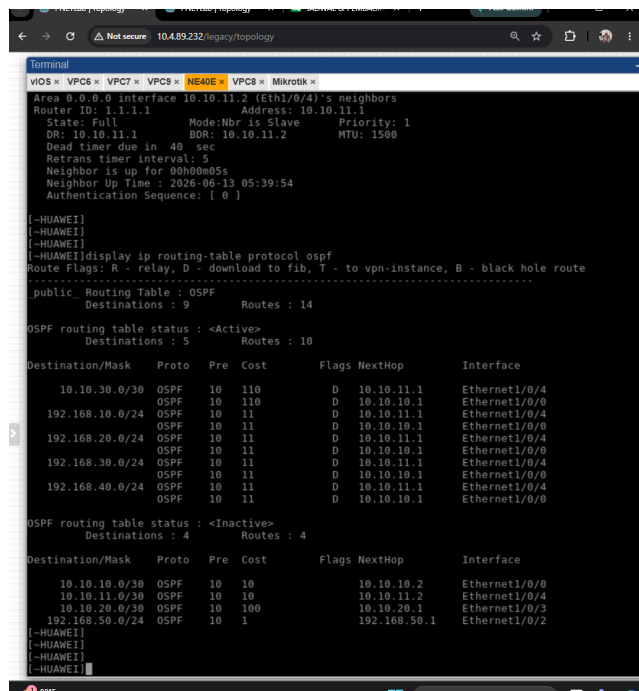
```
1 display ospf peer
2 display ip routing-table protocol ospf
3 display ospf routing
```



```
Terminal
viOS > VPC6 > VPC7 > VPC8 > NE40E > VPC8 > Mikrotik >
undo shutdown
dcn
dcn mode vlan
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0]quit
[*HUAWEI-ospf-1]quit
[*HUAWEI]interface Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0]description LINK1-TO-CISCO
[*HUAWEI-Ethernet1/0/0]ospf cost 10
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4]description LINK2-TO-CISCO
[*HUAWEI-Ethernet1/0/4]ospf cost 10
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3]description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ospf cost 100
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
[*HUAWEI]display ospf peer
(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: None MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m05s
Neighbor Up Time : 2026-06-13 05:39:54
Authentication Sequence: [ 0 ]
Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 40 sec
Retrans timer interval: 5
... More ...
```

Gambar 28



```
Terminal
viOS > VPC6 > VPC7 > VPC8 > NE40E > VPC8 > Mikrotik >
Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00h00m05s
Neighbor Up Time : 2026-06-13 05:39:54
Authentication Sequence: [ 0 ]

[*HUAWEI]
[*HUAWEI]
[*HUAWEI]
[*HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9 Routes : 14

OSPF routing table status : <Active>
Destinations : 5 Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
10.10.10.0/24 OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.30.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.40.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2

[*HUAWEI]
[*HUAWEI]
[*HUAWEI]
```

Gambar 29

```

Terminal
VLOS * VPC6 * VPC7 * VPC8 * NE40E * VPC8 * Mikrotik *
authentication-scheme default
authentication-mode local radius
#
authorization-scheme default
#
accounting-scheme default0
#
accounting-scheme default1
#
domain default0
#
[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2         0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2         0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2         0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1         0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1         0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1         0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1         0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1         0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1         0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2         0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
128.1.138.137/32  0         Direct    128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]S

```

Gambar 30

5. Konfigurasi OSPF pada Mikrotik

```
1 /routing ospf instance set default router-id=3.3.3.3
```

Tambahkan network:

```
1 /routing ospf network add network=10.10.20.0/30 area=backbone
2 /routing ospf network add network=10.10.30.0/30 area=backbone
```

Atur Cost:

```
1 /routing ospf interface add interface=ether1 cost=100
2 /routing ospf interface add interface=ether2 cost=100
```

Verifikasi:

```
1 /routing ospf neighbor print
2 /ip route print where ospf
```

```

vIOS < VPC6 < VPC7 < VPC8 < NE40E < VPC8 < Mikrotik <
2 10.10.20.1 56 255 1ms
3 10.10.20.1 56 255 1ms
4 10.10.20.1 56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@Mikrotik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 1ms
1 10.10.20.1 56 255 1ms
sent=2 received=2 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=1ms

[admin@Mikrotik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 1ms
2 10.10.30.1 56 255 1ms
3 10.10.30.1 56 255 1ms
4 10.10.30.1 56 255 2ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@Mikrotik] > /routing ospf instance set default router-id=3.3.3.3
[admin@Mikrotik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@Mikrotik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@Mikrotik] > /routing ospf interface add interface=ether1 cost=100
[admin@Mikrotik] > /routing ospf interface add interface=ether2 cost=100
[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=2 ls-requests=0 db-summaries=0 adjacency=6s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=1 ls-requests=0 db-summaries=0 adjacency=12s
[admin@Mikrotik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.30.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.30.1 110
[admin@Mikrotik] >

```

Gambar 31

6. Verifikasi OSPF Neighbor

Cisco Router:

```
1 show ip ospf neighbor
```

Hasil yang diharapkan:

- Neighbor Huawei muncul dua kali.
- Neighbor MikroTik muncul satu kali.

```

← → 🔒 Not secure 10.4.89.232/legacy/topology 🔍 ☆ 📄 🔄 ⓘ
Terminal
vIOS < VPC6 < VPC7 < VPC8 < NE40E < VPC8 < Mikrotik <
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 4)

Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 1.1.1.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
10.10.10.0 0.0.0.3 area 0
10.10.11.0 0.0.0.3 area 0
10.10.30.0 0.0.0.3 area 0
192.168.10.0 0.0.0.255 area 0
192.168.20.0 0.0.0.255 area 0
192.168.30.0 0.0.0.255 area 0
192.168.40.0 0.0.0.255 area 0
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 110)

Router#
Router#
Router#
Router#
Router#show ip ospf neighbor
Router#
*Jun 13 05:39:32.107: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING
ULL, Loading Done
*Jun 13 05:39:32.107: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING
ULL, Loading Done
*Jun 13 05:42:20.507: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING
ULL, Loading Done
*Jun 13 05:43:00.827: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING
ULL, Loading Done
Router#
Router#
Router#
Router#
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/DR 00:00:34 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/DR 00:00:35 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:37 10.10.10.2 GigabitEthernet0/3
Router#

```

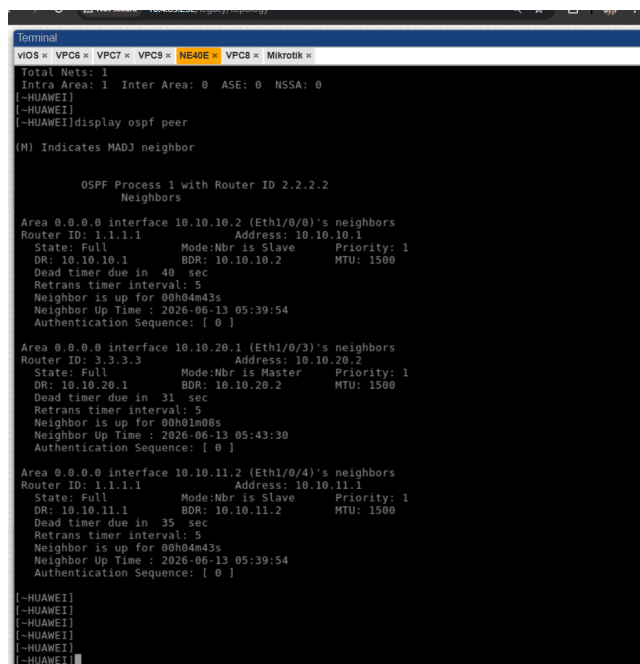
Gambar 32

Huawei Router:

```
1 display ospf peer
```

Hasil yang diharapkan:

- Peer OSPF muncul dua kali.
- Peer MikroTik muncul satu kali.



```
Terminal
VLOS * VPC6 * VPC7 * VPC9 * NE40E * VPC8 * Mikrotik *
Total Neighbors: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
(-HUAWEI)
(-HUAWEI)display ospf peer
(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00h04m43s
Neighbor Up Time : 2026-06-13 05:39:54
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 31 sec
Retrans timer interval: 5
Neighbor is up for 00h01m08s
Neighbor Up Time : 2026-06-13 05:43:30
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00h04m43s
Neighbor Up Time : 2026-06-13 05:39:54
Authentication Sequence: [ 0 ]

(-HUAWEI)
(-HUAWEI)
(-HUAWEI)
(-HUAWEI)
(-HUAWEI)
```

Gambar 33

MikroTik:

```
1 /routing ospf neighbor print
```

Hasil yang diharapkan:

- Neighbor Cisco muncul.
- Neighbor Huawei muncul.



Cisco Router:

```
1 show ip route 192.168.50.0
2 show ip route ospf
```



```
1 display-ip routing-table protocol ospf
```

```
Terminal
vIOS < VPC6 < VPC7 < VPC8 < NE40E < VPC8 < Mikrotik <
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h06m49s
Neighbor Up Time : 2026-06-13 05:39:54
Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9 Routes : 14

OSPF routing table status : <Active>
Destinations : 5 Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
10.10.30.0/30 OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.30.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.40.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
```

Gambar 36

MikroTik:

```
1 /ip route print where ospf
```

```
Terminal
vIOS < VPC6 < VPC7 < VPC8 < NE40E < VPC8 < Mikrotik <
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h06m49s
Neighbor Up Time : 2026-06-13 05:39:54
Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9 Routes : 14

OSPF routing table status : <Active>
Destinations : 5 Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
10.10.30.0/30 OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.30.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.40.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
```

Gambar 37

1.4 8. Verifikasi Koneksi End-to-End

Dari PC VLAN 10:

```
1 ping 192.168.50.10
```

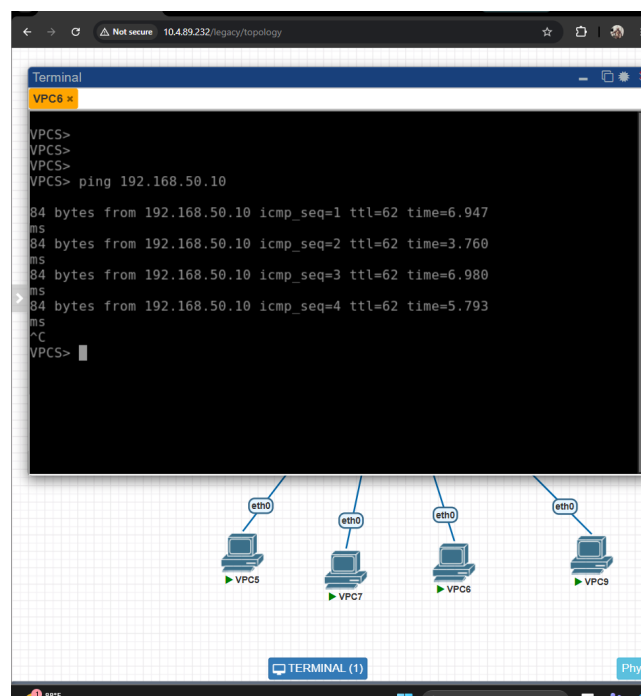
```
terminal
VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 192.168.10.105/24
GATEWAY   : 192.168.10.1
DNS       : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE : 86396, 86400/43200/75600
MAC       : 00:50:79:66:68:d6
I/POR     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.956 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.672 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.913 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=2.021 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=2.997 ms
^C
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=4.934 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=3.288 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=2.285 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.926 ms
^C
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=6.634 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.129 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=3.229 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.403 ms
^C
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=6.718 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=3.243 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=4.814 ms
^C
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.508 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=9.023 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.482 ms
^C
VPCS>
```

Gambar 38

Dari PC VLAN 20:

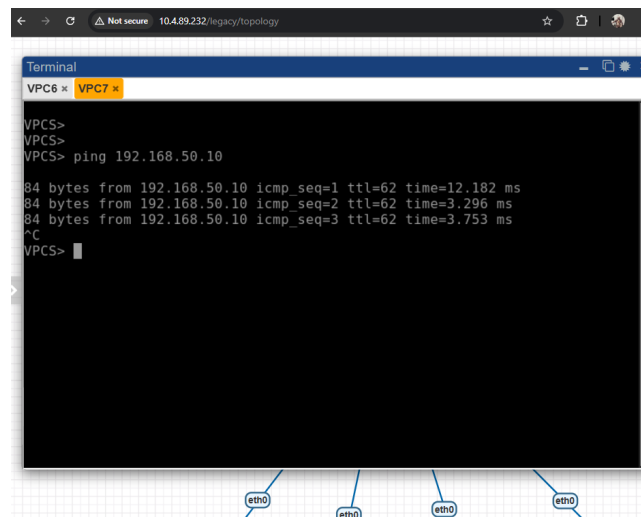
```
1 ping 192.168.50.10
```



Gambar 39

Dari PC VLAN 30:

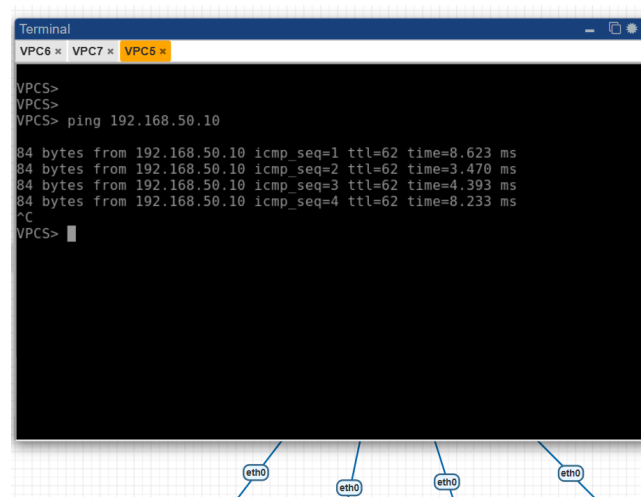
```
1 ping 192.168.50.10
```



Gambar 40

Dari PC VLAN 40:

```
1 ping 192.168.50.10
```



Gambar 41

Dari PC LAN Huawei:

```

1 ping 192.168.10.11
2 ping 192.168.20.11
3 ping 192.168.30.10
4 ping 192.168.40.10

```



```
Terminal
VPC6 x VPC7 x VPC8 x VPC8 x vIOS x
192.168.10.11 icmp_seq=4 timeout
192.168.10.11 icmp_seq=5 timeout

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=3.806 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=4.274 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=4.193 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=4.480 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=4.302 ms
^C
VPCS> ping 192.168.10.11

192.168.10.11 icmp_seq=1 timeout
192.168.10.11 icmp_seq=2 timeout
^C
VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=7.408 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=5.619 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=3.547 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=4.587 ms
^C
VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=9.290 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=3.570 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=5.444 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=4.719 ms
^C
VPCS>
```

Gambar 42

Praktikum 5 Pengujian Failover OSPF

1. Kondisi Normal: Dua Link Cisco-Huawei Aktif

Cek route dari Cisco menuju LAN Huawei:

```
1 show ip route 192.168.50.0
```

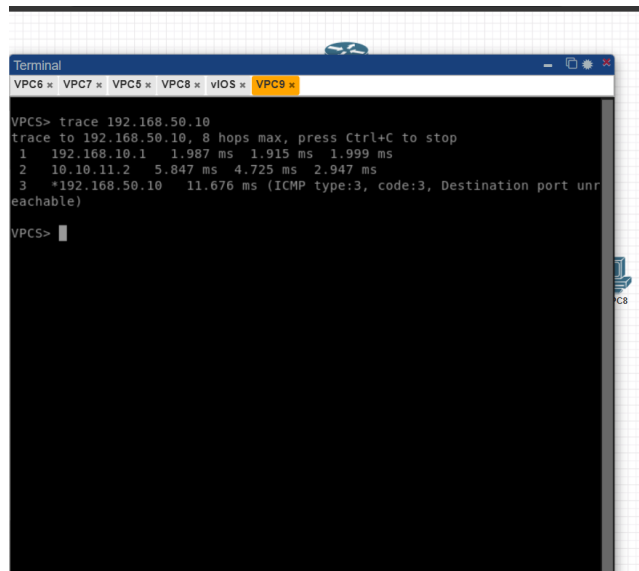
```
10.4.89.232/legacy/topology
Terminal
VPC6 x VPC7 x VPC8 x VPC8 x vIOS x
192.168.10.0/24 is directly connected, GigabitEthernet0/2.10
192.168.10.1/32 is directly connected, GigabitEthernet0/2.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
192.168.20.0/24 is directly connected, GigabitEthernet0/2.20
192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
192.168.30.0/24 is directly connected, GigabitEthernet0/2.30
192.168.30.1/32 is directly connected, GigabitEthernet0/2.30
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
192.168.40.0/24 is directly connected, GigabitEthernet0/2.40
192.168.40.1/32 is directly connected, GigabitEthernet0/2.40
192.168.50.0/24 [110/11] via 10.10.11.2, 00:16:08, GigabitEthernet0/
[110/11] via 10.10.10.2, 00:15:58, GigabitEthernet0/

router#
router#
router#
router#
router#
router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.10.2 on GigabitEthernet0/3, 00:16:10 ago
Routing Descriptor Blocks:
* 10.10.11.2, from 2.2.2.2, 00:16:20 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
10.10.10.2, from 2.2.2.2, 00:16:10 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
router#
```

Gambar 43

Lakukan trace dari PC VLAN menuju LAN Huawei:

```
1 trace 192.168.50.10
```



Gambar 44

Hasil yang diharapkan:

Traffic melewati jalur langsung Cisco - Huawei.

2. Kondisi 1 Link Cisco-Huawei Mati

Matikan salah satu link Cisco-Huawei.

```

1 configure terminal
2 interface gi0/3
3 shutdown
4 exit

```

Cek route dari Cisco menuju LAN Huawei:

```

1 show ip route 192.168.50.0

```

Lakukan trace dari PC VLAN menuju LAN Huawei:

```

1 trace 192.168.50.10

```

Hasil yang diharapkan:

Traffic tetap melewati jalur langsung Cisco.

```
Terminal
VPC6 x VPC7 x VPC8 x vIOS x VPC9 x
Router(config)#
*Jun 13 05:57:40.367: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Interface down or detached
*Jun 13 05:57:42.338: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun 13 05:57:43.338: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
Router(config)#
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
*Jun 13 05:57:55.906: %GRUB-5-CONFIG WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun 13 05:57:56.616: %GRUB-5-CONFIG WRITTEN: GRUB configuration was written to disk successfully.
Router(config)#end
Router#
*Jun 13 05:58:02.349: %SYS-5-CONFIG I: Configured from console by console
Router#
% Type "show ?" for a list of subcommands
Router#
Router#
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:18:28 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:18:38 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#
```

Gambar 45

```
Terminal
VPC6 x VPC7 x VPC8 x vIOS x VPC9 x
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    1.987 ms  1.915 ms  1.999 ms
 2  10.10.11.2     5.847 ms  4.725 ms  2.947 ms
 3  *192.168.50.10 11.676 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.340 ms  2.037 ms  2.833 ms
 2  10.10.11.2     4.017 ms  5.057 ms  5.321 ms
 3  *192.168.50.10 8.055 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> s
```

Gambar 46

1.5 3. Kondisi Dua Link Cisco-Huawei Mati

Matikan link Cisco-Huawei yang kedua:

```
1 configure terminal
2 interface gi0/0
3 shutdown
4 exit
```

Cek route dari Cisco menuju LAN Huawei:

```
1 show ip route 192.168.50.0
```

Lakukan trace dari PC VLAN menuju LAN Huawei:

```
1 trace 192.168.50.10
```

Hasil yang diharapkan:
Route berpindah ke jalur backup melalui MikroTik.
Jalur:

PC VLAN → Cisco Router → MikroTik → Huawei Router → LAN Huawei

4. Mengaktifkan Kembali Dua Link Cisco-Huawei

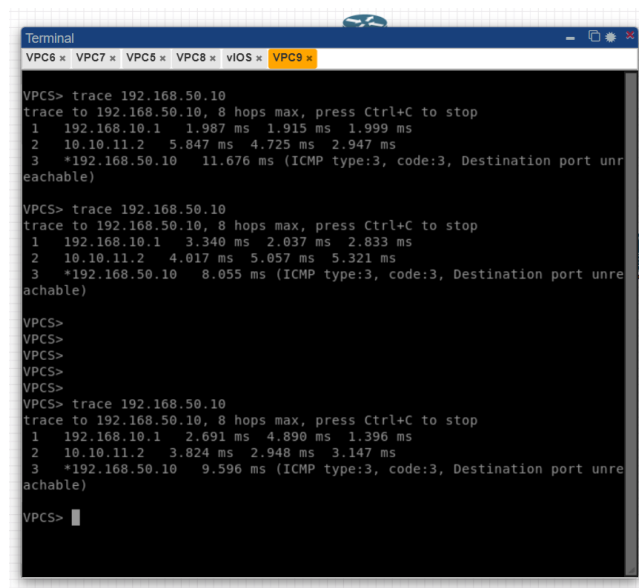
```
1 configure terminal
2 interface gi0/3
3   no shutdown
4 exit
5
6 interface gi0/0
7   no shutdown
8 exit
```

Cek route dari Cisco menuju LAN Huawei:

```
1 show ip route 192.168.50.0
```

Lakukan trace dari PC VLAN menuju LAN Huawei:

```
1 trace 192.168.50.10
```



Gambar 47

Hasil yang diharapkan:
Jalur routing kembali pindah ke link utama Cisco-Huawei.

2 Analisis Hasil Percobaan

Praktikum ini berfokus pada penerapan konsep jaringan komputer mulai dari segmentasi jaringan menggunakan VLAN, konfigurasi gateway pada router, pengalamatan IP antarperangkat jaringan,

implementasi routing dinamis OSPF pada perangkat dari vendor yang berbeda, hingga pengujian kemampuan jaringan dalam mempertahankan konektivitas saat terjadi kegagalan jalur. Berdasarkan seluruh rangkaian kegiatan yang telah dilakukan, tujuan praktikum dapat tercapai dengan baik dan sebagian besar konfigurasi berjalan sesuai perencanaan.

Pada sesi pertama, dilakukan pembentukan beberapa VLAN pada Cisco Switch, yaitu VLAN 10, 20, 30, dan 40. Setiap port yang terhubung ke perangkat akhir ditempatkan pada VLAN yang sesuai, sementara koneksi antara switch dan router dikonfigurasi sebagai trunk agar dapat membawa lalu lintas dari seluruh VLAN. Hasil pemeriksaan konfigurasi menunjukkan bahwa segmentasi jaringan telah terbentuk dengan benar dan mekanisme trunking berfungsi sebagaimana mestinya.

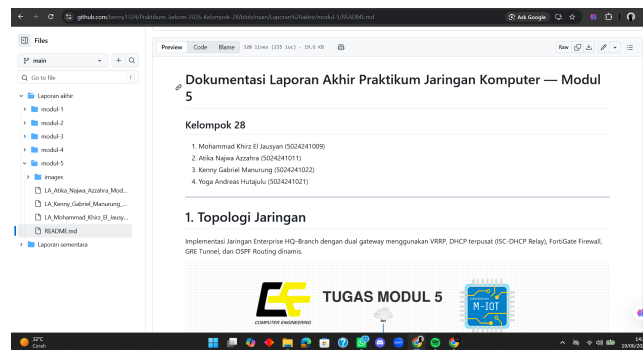
Tahap berikutnya berfokus pada konfigurasi Cisco Router sebagai pusat komunikasi bagi seluruh VLAN. Selain mengaktifkan fungsi gateway, router juga dikonfigurasi untuk menyediakan layanan DHCP pada VLAN 10 dan VLAN 20 sehingga klien dapat memperoleh alamat IP secara otomatis. Adapun perangkat yang berada pada VLAN 30 dan VLAN 40 menggunakan pengaturan alamat IP secara manual. Pengujian yang dilakukan menunjukkan bahwa perangkat pada segmen jaringan yang berbeda tetap dapat saling berkomunikasi, menandakan bahwa proses inter-VLAN routing telah berjalan dengan sukses.

Pada praktikum selanjutnya, dilakukan pengaturan alamat IP pada setiap antarmuka yang menghubungkan Cisco Router, Huawei Router, dan MikroTik. Jaringan lokal yang terhubung ke Huawei Router juga berhasil diimplementasikan sesuai topologi yang diberikan. Berdasarkan hasil uji konektivitas antarrouter, seluruh perangkat dapat saling menjangkau tanpa adanya kendala komunikasi. Kondisi tersebut menunjukkan bahwa konfigurasi addressing telah dilakukan secara tepat dan semua interface yang digunakan beroperasi dengan normal.

Implementasi protokol OSPF dilakukan pada tahap berikutnya dengan tujuan membangun sistem routing dinamis antarrouter yang berasal dari vendor berbeda. Setelah konfigurasi selesai diterapkan, setiap router berhasil membentuk hubungan tetangga (neighbor) OSPF dan melakukan pertukaran informasi rute. Tabel routing pada masing-masing perangkat juga telah terisi dengan jaringan yang diperoleh dari router lain sehingga proses pengiriman paket dapat dilakukan secara otomatis. Selama pengujian konektivitas ditemukan satu kondisi ketika komputer pada jaringan Huawei mengalami timeout saat mengakses alamat 192.168.10.11. Namun setelah dilakukan verifikasi, tidak ditemukan kesalahan konfigurasi baik pada OSPF maupun tabel routing. Kemungkinan besar kondisi tersebut disebabkan oleh proses penyesuaian rute selama konvergensi atau respons awal perangkat tujuan yang terlambat. Selain kejadian tersebut, seluruh pengujian komunikasi antarjaringan memberikan hasil yang sesuai harapan.

Tahap terakhir ditujukan untuk mengamati mekanisme failover pada OSPF. Pengujian dilakukan dengan memutus koneksi utama antara Cisco Router dan Huawei Router. Ketika salah satu jalur utama tidak tersedia, lalu lintas jaringan masih dapat diteruskan melalui jalur utama lainnya tanpa kehilangan konektivitas. Saat kedua jalur utama dinonaktifkan, OSPF secara otomatis memilih rute alternatif melalui MikroTik sebagai jalur cadangan. Setelah koneksi utama dipulihkan, proses routing kembali menggunakan jalur dengan prioritas tertinggi. Hasil ini menunjukkan bahwa OSPF mampu melakukan adaptasi terhadap perubahan topologi jaringan secara otomatis. Walaupun demikian, terdapat kekurangan pada aspek dokumentasi karena beberapa bukti hasil pengujian trace tidak berhasil disertakan sesuai dengan ketentuan yang tercantum dalam modul praktikum.

3 Hasil Tugas Modul



Gambar 48

4 Kesimpulan

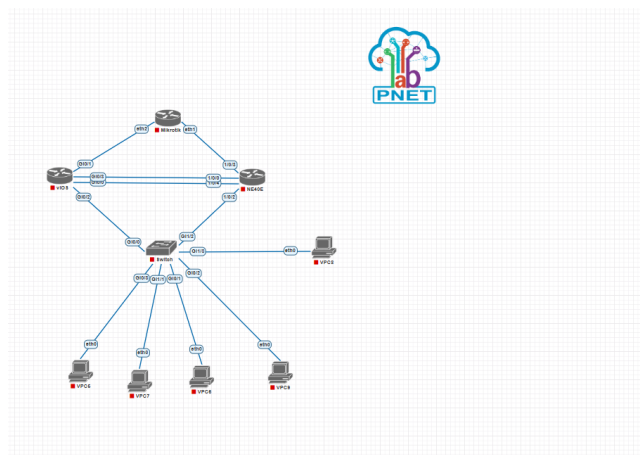
Dari seluruh rangkaian praktikum yang telah dilaksanakan, dapat diketahui bahwa proses perancangan dan implementasi jaringan berhasil direalisasikan menggunakan perangkat Cisco Switch, Cisco Router, Huawei Router, dan MikroTik. Segmentasi jaringan melalui VLAN dapat diterapkan dengan baik pada switch, sedangkan Cisco Router mampu menjalankan fungsi inter-VLAN routing menggunakan metode Router-on-a-Stick sehingga komunikasi antarsegmen jaringan dapat berlangsung tanpa hambatan.

Konfigurasi pengalamatan IP pada setiap koneksi antarrouter juga berhasil dilakukan sesuai dengan topologi yang dirancang. Dengan konfigurasi tersebut, seluruh perangkat jaringan dapat saling terhubung dan melakukan pertukaran data. Selain itu, penerapan protokol routing dinamis OSPF pada perangkat yang berasal dari vendor berbeda menunjukkan hasil yang memuaskan. Setiap router mampu membangun hubungan neighbor serta mendistribusikan informasi rute secara otomatis, sehingga kebutuhan akan konfigurasi routing statis dapat diminimalkan.

Hasil pengujian ketahanan jaringan menunjukkan bahwa OSPF memiliki kemampuan untuk beradaptasi terhadap perubahan topologi. Ketika jalur utama mengalami kegagalan, proses pemilihan rute alternatif dapat dilakukan secara otomatis tanpa memutuskan komunikasi antarjaringan. Walaupun ditemukan satu kasus timeout pada saat pengujian konektivitas dari jaringan Huawei menuju alamat 192.168.10.11 dan masih terdapat beberapa dokumentasi pengujian yang belum lengkap, kondisi tersebut tidak memengaruhi keberhasilan implementasi jaringan secara keseluruhan. Seluruh fungsi utama yang menjadi target praktikum tetap dapat dijalankan dengan baik.

Melalui kegiatan praktikum ini, praktikan memperoleh pengalaman langsung dalam membangun jaringan berbasis VLAN, menghubungkan beberapa VLAN menggunakan routing, melakukan konfigurasi pengalamatan IP pada perangkat jaringan, menerapkan protokol OSPF pada lingkungan multi-vendor, serta memahami mekanisme redundansi dan failover yang berperan penting dalam menjaga ketersediaan layanan jaringan.

5 Lampiran



Gambar 49: Topologi praktikum